

Question 1(a) [22 marks available]

For each test of the program for 1(a) you need to type in a word of 1 to 10 uppercase letters. The output should be a pair of integers. Both integers in the output need to be correct and in order for marks to be scored.

Tests *must* terminate in 1 second to receive marks.

[1] BIO	45 118
[2] M	3 3
[2] EE	25 25
[2] QN	9 8
[2] AEY	25 121
[2] UPKF	1 39
[2] TSRQP	2931 782
[2] JNRVCGK	73306 53868
[2] OLYMPIAD	341954 188975
[2] ABCDEFGHIJ	606445 9764844

Additional marks are available for general program behaviour:

- [1] Program inputs a word.
- [1] For each test two numbers are output.
- [1] All tests terminate without crashing / hanging.

Question 1(b) [3 marks available]

- [1] QIGS
- [2] AGMSTKGMSO

Question 1(c) [5 marks available]

- [5] 35176

Question 2(a) [25 marks available]

There are 11 tests used to check 2(a). For each test you will need to type in three integers and an uppercase word.

For each test you should see an uppercase word, which needs to match exactly to score marks.

Tests *must* terminate in 1 second to receive marks.

[1] 6 3 2 BB	BD
[2] 26 0 0 M	M
[2] 26 2 0 P	W
[2] 26 1 1 R	E
[2] 5 10 10 CAB	CBE
[2] 6 10 10 FADE	ABFF
[2] 9 30 2 ABCDE	AHICD
[3] 10 30 26 BADGE	FFFEH
[3] 19 2026 3 GARDNER	NJFRAEG
[3] 26 5000 51 SQUEAMISH	MXBMVODHN
[3] 20 1000 4 OSSIFRAGE	DBOLRACBT

Question 2(b) [2 marks available]

- [2] LOUDER

Question 2(c) [4 marks available]

Either of the following words can be given to score all [4] marks. (They only differ in the positions of the A and B characters.)

- [4] QBDPOACNMHLKGFJIE
- QADPOBCNMHLKGFJIE

Question 2(d) [5 marks available]

- [5] 230

Question 3(a) [24 marks available]

Each test for 3(a) you will need to enter a single integer and should see a single integer output.

There are no marks for incorrect answers, and tests *must* terminate in 1 second to receive marks.

[1]	99	45
[2]	9	0
[2]	56	15
[2]	6000	3756
[2]	345678	84835
[2]	23456780	1120614
[2]	1234567890	8330472
[2]	112233445566	57604755
[3]	81020268102026	803205280
[3]	12345678987654321	5413024975
[3]	8877665544332211999	35167216965

Question 3(b) [2 marks available]

[2] 8,111,111

Question 3(c) [4 marks available]

[4] 112,444,777,888,822,225

Question 3(d) [4 marks available]

[4] 7974